

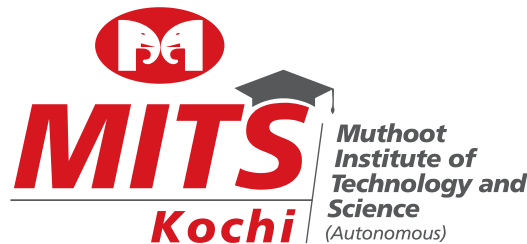
LucidCare: An Ensemble Patient Centric Clinical Report Interpretation Using Explainable AI

PROJECT PHASE 2
ADD415

*Project Phase 2 Report submitted in partial fulfillment for the degree
of
B.Tech Artificial Intelligence and Data Science*

Submitted by

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DECLARATION

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously or written by another person nor material which has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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CERTIFICATE

This is to certify that the Project Phase 2 report entitled “**LucidCare: An Ensemble Patient Centric Clinical Report Interpretation Using Explainable AI**” submitted by **ABHISHEK BIJU (Reg No: MUT22AD005), ALINA PINHEIRO (Reg No: MUT22AD014), JOEL S MATHEWS (Reg No: MUT22AD043), ROSHEN REJI (Reg No: MUT22AD062)** of Semester VII is a bonafide account of the work done by them under our supervision during the academic year 2025-26.

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Abstract

Understanding such complex medical information, like radiographic reports, is difficult to attain by persons without medical backgrounds, therefore there will be always a dependency on healthcare professionals and delayed interpretation. The proposed project presents an AI-powered system for generating simple, patient-friendly medical reports from X-ray images enriched by insights of explainable AI. It integrates a vision-based model for image analysis, a classifier that detects and categorizes medical conditions, and a large language model for generating human-understandable explanations clearly. A key novelty in the work is the bidirectional interaction between the large language model and the vision model that allows dynamic reasoning, highly accurate abnormality detection, and longitudinal comparison of current and past reports. The generated reports are in Malayalam, making the information more accessible to the local user. Further, this system helps to promote health literacy among patients by translating complex medical jargon into easily understandable language and empowers the patient for active participation in self-care.

CO–PO/PSO Articulation Matrix and SDG Alignment

Course Outcome	Program Outcome (PO)											Program Specific Outcome (PSO)	
	1	2	3	4	5	6	7	8	9	10	11	1	2
CO1: Model and solve real world problems by applying knowledge across domains	2	2	2	1	2	2	1	1	1	1	2	2	2
CO2: Develop products, processes or technologies for sustainable and socially relevant applications	2	2	2		1	3	3	1		1	1	2	2
CO3: Function effectively as an individual and as a leader in diverse teams and to comprehend and execute designated tasks								3	2	2	1		
CO4: Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms					2		3	2	2	3	2		
CO5: Identify technology/research gaps and propose innovative/creative solutions	2	3	3	1	2						1	2	2
CO6: Organize and communicate technical and scientific findings effectively in written and oral forms					2		2	2	3	1	1	1	1

Figure 1: CO–PO/PSO Articulation Matrix

Table 1: Alignment of Project with Sustainable Development Goals

SDG	Sustainable Development Goal	Project Alignment
3	Good Health and Well-being	The project's primary aim is to improve health outcomes by making complex medical reports understandable.
9	Industry, Innovation and Infrastructure	An AI-driven solution integrating multimodal systems (XAI, YOLOv8) for practical clinical innovation.
10	Reduced Inequalities	Addresses health information inequality by bridging the communication gap with bilingual support.
4	Quality Education	Supports health literacy by providing simplified medical explanations, empowering patients with knowledge.

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List of Abbreviations

YOLO	You Only Look Once
SVM	Support Vector Machine
LLM	Large Language Model
VLM	Vison-Language Model
LIME	Local Interpretable Model-Agnostic Explanations
CNN	Convolutional Neural Network
GAN	Generative Adversarial Network
XAI	Explainable AI

Chapter 1

Introduction

1.1 Motivation

Medical reports, particularly radiology and discharge summaries, are frequently written in complex, technical language that is difficult for the average patient to understand. This has brought about a communication gap in the form of highly specialized jargon between healthcare providers and their patients. This will serve to produce a lesser active involvement of the patient and an informed participation by them in their own health decisions.

While the existing systems try to solve this by using LLMs to convert all medical terminology into some light, easy-to-understand summaries, they have serious limitations. Most of the solutions involve textual data only; they either don't incorporate vital insights from the medical images, like X-rays, or simply do not analyze these elements, which may form the backbone of a certain diagnosis of a patient. This textual approach often confines the system to providing a holistic and accurate explanation, combining text and visual information.

Besides, most of these existing systems provide their output in simplified form only in English. This is a big disadvantage for those patients of regional areas who prefer or need information of a critical nature to be in their native language, further hindering accessibility and comprehension.

1.2 Problem Statement

With these challenges, the need for a more comprehensive AI-driven solution is rather outspoken. The aim is to develop a multimodal AI system that will simplify even the most complex medical reports and integrate into the analysis of medical images, such as X-rays.

This proposed system must overcome the limitations of existing solutions by more user-friendly approach. It analyzes both textual reports and radiological images to provide a complete diagnostic summary. Generates simplified, accurate, and easy-to-understand explanations of the findings and also providing bilingual support, specifically in English and Malayalam, to make critical medical information accessible and understandable for regional language speakers. Aim is improve patient accessibility and understanding, empowering local patients to be more involved in their own care.

Chapter 2

Literature Review

2.1 Patient-Centered Reports with Generative AI

Park et al. (2024) conducted a comprehensive investigation into the application of generative artificial intelligence for producing patient-centered radiology reports, focusing specifically on spine MRI findings. Their study examined a dataset comprising 685 authentic spine MRI reports, from which generative AI models were used to create three distinct types of outputs: a concise summary report, a patient-friendly report written in accessible language, and personalized clinical recommendations derived from the imaging results.

The researchers employed a systematic evaluation framework to assess the accuracy, readability, and clinical reliability of each AI-generated output. One of the most striking outcomes of the study was the marked improvement in patient comprehension levels — rising from an average score of 2.71 for conventional radiology reports to 4.69 for AI-generated patient-friendly versions on a standardized 5-point understanding scale. This finding underscores the substantial potential of generative AI to bridge the communication gap between radiologists and patients, transforming complex medical jargon into clear, comprehensible narratives.

Despite these advancements, the study also identified certain limitations inherent to generative AI. Specifically, minor hallucinations—defined as fabricated or inaccurate statements—were observed in approximately 1.12% of outputs, while potentially harmful translations or misinterpretations occurred in 7.40% of cases. Nevertheless, these error rates were considered minimal relative to the overall improvements in interpretability and accessibility.

Ultimately, Park et al. concluded that generative AI significantly enhances the quality and clarity of radiology reports, streamlines report-generation workflows, and aligns with a broader shift toward patient-centered communication in radiology practice. The study presents compelling evidence that, when properly validated and monitored, AI-driven reporting systems can serve as powerful tools for improving both the efficiency and the human-centeredness of medical imaging communication.

2.2 Explainable AI for Pulmonary Disease Interpretation

Naz et al. (2023) present a novel framework that goes beyond mere classification of pulmonary diseases from chest radiographs, emphasizing interpretability through explainable AI (XAI). Their methodology leverages a Convolutional Neural Network (CNN) with transfer learning to accurately identify a range of pulmonary conditions, including edema, tuberculosis, nodules, and pneumonia, achieving a high classification accuracy of 97%.

What distinguishes this work is its integration of an interpretable explanation module. Using LIME (Local Interpretable Model-Agnostic Explanations), the system highlights the specific regions of the radiograph that contributed most to each diagnostic decision. These AI-generated attention regions were rigorously validated against the markings provided by expert human radiologists, demonstrating a strong alignment between the model’s reasoning and human clinical judgment.

This study is particularly foundational for our project, as it establishes a proven methodology for incorporating XAI into medical imaging systems. The combination of high diagnostic accuracy with clear, human-validated explanations provides a blueprint for the “LUCIDCARE” system, ensuring that our AI not only predicts disease effectively but also remains transparent and interpretable, a core requirement for clinical trust and adoption.

2.3 Vision-Language Models for Chest Radiograph Descriptions

Ji et al. (2024) investigate the automatic generation of textual descriptions from chest radiographs using a vision-language model, addressing a central challenge in radiology AI: translating complex visual data into coherent, clinically relevant text. Their study demonstrates that vision-language models can effectively bridge the gap between raw image pixels and meaningful semantic descriptions, enabling automated generation of radiology report drafts.

By systematically evaluating the model’s performance, Ji et al. provide empirical evidence that such architectures are capable of capturing key radiographic features and expressing them in descriptive text, closely aligning with the workflow of human radiologists. This work serves as a foundational baseline for our system, supporting our approach of generating structured outputs (in JSON format) from the vision model, which can then be interpreted and expanded by a large language model (LLM) to produce patient- or clinician-facing reports.

In essence, Ji et al. establish the feasibility and reliability of vision-to-text translation in medical imaging, reinforcing the rationale for our model’s design that separates visual feature extraction from textual report generation.

2.4 A Vision-Language Foundation Model for Report Generation

Bluthgen et al. (2024) present a large-scale vision-language foundation model tailored for the automatic generation of radiology reports, representing a state-of-the-art approach in the field. Their model is trained on an extensive dataset of image-report pairs, enabling it to learn the intricate

relationships between visual abnormalities in medical images and the precise clinical terminology used to describe them.

The study demonstrates that such foundation models can capture complex, multimodal associations, producing high-quality, contextually accurate radiology reports with minimal human intervention. By integrating both visual and textual modalities at scale, the model exemplifies the potential of large AI systems to synthesize information across modalities, ensuring that the generated outputs are both clinically relevant and linguistically coherent.

This work establishes a crucial precedent for our project, supporting our objective of leveraging a powerful large language model (LLM) to consolidate insights from both the vision and text components of our system. Bluethgen et al. provide empirical evidence that multimodal foundation models can serve as robust, scalable solutions for high-fidelity radiology report generation, directly informing the design principles of our own pipeline.

2.5 Review of Vision-Language Models for Medical Applications

Hartsock and Rasool (2024) present a comprehensive review of recent advancements in medical vision-language models (VLMs), with a particular focus on their applications in radiology report generation and visual question answering (VQA). The paper systematically examines the architectures and pre-training strategies of 16 state-of-the-art medical VLMs and provides an overview of 18 publicly available medical VLM datasets, offering a broad perspective on the current landscape of multimodal medical AI.

A key insight from their analysis is the prevalent use of Transformer-based architectures to effectively fuse visual and textual data, highlighting a consistent trend toward models capable of capturing complex relationships between medical images and their semantic descriptions. The review also identifies ongoing challenges in the field, including limited availability of high-quality annotated data and the lack of standardized evaluation metrics to assess model performance comprehensively.

This review is particularly relevant to our project as it validates our choice of a Transformer-based multimodal architecture and provides critical context for the current limitations and opportunities in medical VLM research. By synthesizing these insights, Hartsock and Rasool offer a strategic foundation for designing robust, scalable, and interpretable vision-language systems in healthcare applications.

2.6 Orchestrating XAI for Multimodal and Longitudinal Data

Pahud de Mortanges et al. (2024) tackle the urgent need for explainable AI (XAI) systems capable of handling both multimodal and longitudinal clinical data, which are essential for effective decision-making in medical workflows. They highlight that, despite significant advancements in AI for medical imaging, the “black box” nature of many AI systems remains a major barrier to clinical

adoption, as clinicians require transparent reasoning to trust automated predictions.

To address this, the authors introduce the concept of an “XAI Orchestrator”, a virtual assistant designed to help clinicians synthesize multimodal inputs, AI predictions, and their corresponding explanations into actionable insights. This system is capable of integrating information across text, images, and time-series data, thereby supporting longitudinal analysis and informed clinical decision-making.

The study is highly relevant to our project, as it directly informs the design of our “Comparative Analysis” module, which aims to provide longitudinal insights by comparing current and prior radiology reports. By drawing from the principles of the XAI Orchestrator, our system can enhance interpretability, improve clinical trust, and enable meaningful tracking of patient progress over time.

2.7 Explainable emphysema detection on chest radiographs with explicit radiologic signs

Çallı et al. (2022) focused on improving the interpretability of deep learning models used for detecting emphysema in chest radiographs. Traditional deep learning approaches often function as black box systems, which limits their acceptance in clinical environments. To address this, the authors incorporated explicit radiologic signs into the model, aligning predictions with clinically recognized patterns.

The proposed method leverages domain knowledge by embedding radiological features directly into the learning process. This allows the model to highlight specific visual cues, such as lung texture abnormalities, that are commonly used by radiologists for diagnosis. As a result, the model does not only predict outcomes but also provides reasoning that aligns with medical understanding.

The study demonstrates that integrating explainability improves both model transparency and clinician trust. This approach is particularly important in healthcare, where decisions must be justified and validated. The work sets a foundation for developing AI systems that are not only accurate but also interpretable and clinically reliable.

2.8 Deep learning improves physician accuracy in chest X ray interpretation

Anderson et al. (2024) examined the role of deep learning models as supportive tools in clinical diagnosis, specifically for chest X ray interpretation. The study evaluated how AI assistance influences physician performance, particularly in identifying abnormalities that may be overlooked in manual analysis.

The results showed that physicians supported by deep learning models achieved significantly higher diagnostic accuracy compared to those working independently. The model provided additional insights and highlighted regions of interest, enabling clinicians to make more informed

decisions. This demonstrates the potential of AI in reducing human error and improving diagnostic consistency.

Importantly, the study emphasizes that AI should not replace clinicians but rather augment their capabilities. The collaboration between human expertise and machine intelligence leads to better outcomes. This work reinforces the importance of designing AI systems that integrate seamlessly into clinical workflows.

2.9 A scoping review on multimodal deep learning in biomedical images and texts

Sun et al. (2023) conducted a comprehensive review of multimodal deep learning techniques that integrate biomedical images with textual data. The study highlights the growing need to combine heterogeneous data sources to capture complex relationships in medical analysis.

The review discusses various fusion strategies used in multimodal learning, including early fusion, late fusion, and hybrid approaches. Each method offers different advantages in terms of feature representation and computational efficiency. The authors emphasize that combining modalities leads to richer and more informative representations compared to single modality approaches.

Despite its advantages, the study identifies several challenges in multimodal learning, such as data heterogeneity, alignment issues, and lack of standardized datasets. The authors suggest that future research should focus on improving interpretability and scalability. This work provides a strong theoretical foundation for developing advanced multimodal models in healthcare.

2.10 RAG RLRC LaySum at BioLaySumm: Integrating retrieval augmented generation and readability control for layman summarization of biomedical texts

Fi et al. (2024) proposed a novel framework for generating simplified summaries of complex biomedical texts using retrieval augmented generation and readability control. The study addresses the challenge of making medical information accessible to non experts without compromising accuracy.

The approach combines retrieval mechanisms with generative models to ensure that the generated summaries are both relevant and factually correct. Additionally, readability control is applied to adjust the complexity of the language, making the content easier to understand for a general audience. This is particularly useful for patient education and public health communication.

The results demonstrate that the proposed method produces high quality summaries that balance accuracy and simplicity. This work expands the application of AI beyond prediction and diagnosis, highlighting its role in knowledge dissemination. It also emphasizes the importance of designing AI systems that cater to both experts and non experts in the healthcare domain.

Chapter 3

Proposed Project

3.1 Objectives

The main goals of the "LUCIDCARE" project in developing a multimodal AI system that integrates an LLM with a Vision Model to analyze medical reports and X-ray images. The goal of this system is to synthesize comprehensible yet simple medical explanations understandable by the patient. The system shall be able to conduct comparison studies, enabling it to compare previous and current outputs in order to detect changes, such as normal versus abnormal findings, or increases/decreases of some medical parameter. It will also provide bilingual support-English and Malayalam-to ensure better access to and comprehension of critical medical information by the speakers of the regional language.

3.2 Proposed Solution

The proposed system is an AI-driven platform that will interpret the most complex data in medicine and translate it into comprehensible information for patients. This may involve the consumption of any medical reports, such as in PDF form, comprising both textual clinical findings and radiological images. Core to this innovation is a truly integrated multimodal system that imagines together the power of Vision AI for image understanding with Large Language Model capability in understanding and creating natural language. It enables the correct interpretation of both the visual and textual components of a given medical report and translates those into coherent, patient-centered explanations.

One of the key features in the system is the XAI techniques that will provide clear, transparent, and interpretable reasoning for each diagnosis or observation. Such a system should be able to outline the rationale for each medical conclusion to enable the patients to understand not only the findings but also the underlying logic and clinical relevance of those findings.

The system is modular and hosted around some interconnected modules. In sequence, these modules have been functioning to process the input reports, analyze text and image data, extract relevant clinical insights from the information, and synthesize them into a clear, patient-friendly

format. Having such a modular architecture ensures comprehensive steps in the transformation of complex medical data into accessible and interpretable outputs so that patients can understand their health information with much more involvement.

3.2.1 Report Extraction

This is the very first module of data ingestion. In the case of feeding a medical report into the system, it will first of all apply the process of OCR to extract with high accuracy all the textual content within the file, structured data, clinical notes, and all sorts of annotations. While doing so, it will concurrently extract any embedded medical images like X-rays using image extraction techniques to make sure that both text and visual information will be captured for further analysis in the subsequent modules.

3.2.2 Vision Model (YOLOv8 + DenseNet121)

This module is the central system of the entire image analysis pipeline in the system, which ensures precise detection and classification of abnormalities within medical images. It has been designed as an ensemble model that brings in the best of both worlds of object detection and deep classification networks to get highly accurate and explainable results.

- **YOLOv8 (Detector):** Given an extracted X-ray image, this module feeds into a YOLOv8 model—a version that acts as a very efficient detector. This scans for the appearance and locations of abnormalities, such as lesions, nodules, or other regions of opacity. This outputs the coordinates of the detected sub-regions in JSON format, describing in a structured format where abnormalities are located. It also crops the image around these regions of interest, thus isolating relevant pathological areas better for more detailed analyses in the next step.
- **DenseNet121 (Classifier):** Each of the cropped regions is subsequently fed into a DenseNet model, which makes the fine-grained classification. The classifier identifies visual patterns in that region that would ascertain the type of disease or abnormality, whether it is Pneumonia, COVID-19, Tuberculosis. The model also outputs the confidence score along with the predicted type of disease, indicating the strength of the confidence in making the prediction. These also get formatted as JSON, so it can easily be integrated with explanation and report generation modules downstream.

3.2.3 Model Integration (LLM)

This module takes on the task of integrating all data into a unified, patient-friendly report. The outputs in JSON format from both the YOLOv8 detector and the DenseNet classifier are combined with the text provided by the OCR module, making for a comprehensive multimodal dataset to represent both visual and textual clinical information.

This integrated data is then given as a structured prompt to a Large Language Model, synthesizing the information so that the medical report summary becomes simplified and patient-centric. The generated report should be clear and easily understandable by explaining the finding and clinical significance of the same. Toward improvement in accessibility, bilingual output is the official language, with the final report presented in both English and Malayalam, making it possible for patients coming from diverse linguistic backgrounds to understand their medical information with ease.

3.2.4 Comparative Analysis

This module is designed to give the overall view of a patient's medical history. It gets activated when previous and current medical summaries exist in the system's database. The module compares these reports with each other for finding and underlining changes in medical conditions over time, such as shifts from normal to abnormal findings, or variations in specific clinical parameters.

According to the result of this comparison, the system generates a clear, patient-friendly summary that outlines progressions, improvements, or deteriorations in health status. It allows patients to easily understand how their condition has changed over time and puts that in a valuable context for ongoing care, facilitating informed discussions with healthcare providers.

3.2.5 Backend & Database

This module handles the system's infrastructure and data management. The backend uses FlaskAPI to process frontend requests, coordinate AI models, and deliver structured results efficiently. SupaDB stores patient data, original reports, and generated analyses, using its flexible document structure to support both textual and JSON data for multimodal and longitudinal analysis.

Chapter 4

Project Design

4.1 System Architecture

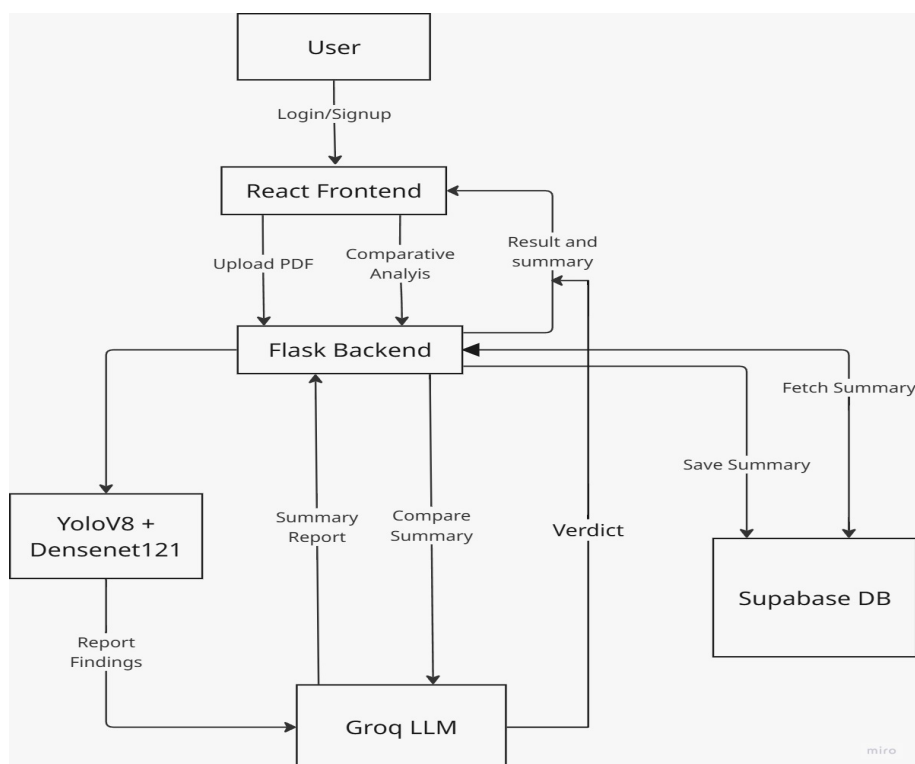


Figure 4.1: Architecture diagram

The proposed system follows a modular full-stack architecture designed for multimodal medical data processing, security, and persistence. The workflow begins at the React Frontend, which serves as the user interface for authentication (Login/Signup) and the primary gateway for data entry. Users can either upload a medical PDF for initial analysis or request a comparative analysis of their historical health data.

The Flask Backend acts as the central orchestration layer, managing the data flow between the

client and various AI services. When a PDF is uploaded, the backend routes the extracted image data to the Vision Model Ensemble, which utilizes YOLOv8 for spatial localization of abnormalities and DenseNet121 for disease classification. The findings from the vision pipeline are then transmitted to the Groq LLM. Simultaneously, the backend extracts textual data from the report and feeds it to the LLM, which synthesizes both modalities into a patient-friendly summary.

For long-term health monitoring, the system integrates a Supabase Cloud Database. The backend performs "Save Summary" and "Fetch Summary" operations to maintain a longitudinal record of patient health. During comparative analysis, historical summaries are retrieved and compared by the Groq LLM to generate a clinical verdict regarding the progression of the condition. The final results, including diagnostic summaries and comparative verdicts, are pushed back to the React Frontend for user visualization.

4.2 Modules

4.2.1 React Frontend

The React Frontend serves as the primary interface for user interaction, designed to provide a seamless and secure experience. This module handles user authentication (Login and Signup) and manages the state of the application. It provides a dedicated portal for patients to upload medical PDF reports and initiate requests for either a new analysis or a longitudinal comparison of past findings. Once the backend processes the data, the frontend is responsible for rendering the complex diagnostic results and AI-generated summaries in an intuitive, patient-friendly dashboard.

4.2.2 Flask Backend

The Flask Backend acts as the central engine of the system, orchestrating the data flow between the client interface and the AI service layers. Upon receiving a PDF upload, this module performs two parallel operations: Text Extraction using libraries like PyPDF2 to capture clinical vitals, and Image Extraction via PyMuPDF (fitz) to isolate radiological scans. It manages the communication with the vision models and the Groq LLM, ensuring that data is correctly formatted and routed. Additionally, it handles the logic for fetching and sorting historical records for the comparative analysis feature.

4.2.3 Vision Ensemble Model (YOLOv8 + DenseNet121)

The Vision Model serves as the core analytical component responsible for processing and interpreting all visual data extracted from the medical report. It functions as an ensemble model composed of two sequential sub-modules.

- **Detector:** This is the first sub-module which receives the radiological image from the Image Extraction process. Its primary task is to identify and localize regions of interest within the image—such as potential lesions, fractures, or abnormal tissue structures. It highlights

specific areas that may indicate medical abnormalities, ensuring that the subsequent stages focus only on the most relevant portions of the image.

- **Classifier:** The second sub-module takes the input from localized regions identified by the Detector. It then performs a classification task to categorize these regions into medically interpretable outcomes, such as the presence or absence of disease, the type of abnormality, or the severity of the detected condition. It leverages deep learning-based image recognition algorithms to ensure high accuracy and reliability in its predictions.

The output of this ensemble consists of "Report Findings," including the condition detected, the confidence score, and the spatial location

4.2.4 Groq LLM

The LLM Model serves as the central integration and generation module within the system architecture. It unifies the analytical outputs from the Vision Ensemble Model with the textual information obtained from the Text Extraction module. Designed to handle multimodal inputs, this component accepts two streams of data — the processed visual insights produced by the Vision Ensemble Model and the raw clinical text extracted from the medical report.

Once both inputs are received, the LLM Model merges image-based findings with contextual textual details. This integration allows the model to cross-reference visual abnormalities with their corresponding clinical descriptions, ensuring that the interpretation is both accurate and contextually meaningful.

After consolidating the multimodal data, the LLM Model generates the final patient-facing output - a simplified and comprehensible version of the medical report. This output translates complex diagnostic information into clear, human-understandable language while preserving the essential medical insights. Through this process, the LLM Model acts as the bridge between advanced medical analysis and patient comprehension, fulfilling the system's goal of making healthcare information more accessible and interpretable.

4.3 Data Flow Diagrams

4.3.1 Level 0



Figure 4.2: Level 0 DFD

The Level 0 Data Flow Diagram, or context diagram, represents the highest-level view of the system's information requirements. The user interacts with the LucidCare system by providing authentication credentials and medical PDF reports as inputs. The system then processes this data and returns a simplified diagnostic Summary to the user. Simultaneously, the system manages data persistence through bidirectional Save/ Fetch operations with the Supabase DB, ensuring that all clinical findings are securely stored for future retrieval.

4.3.2 Level 1

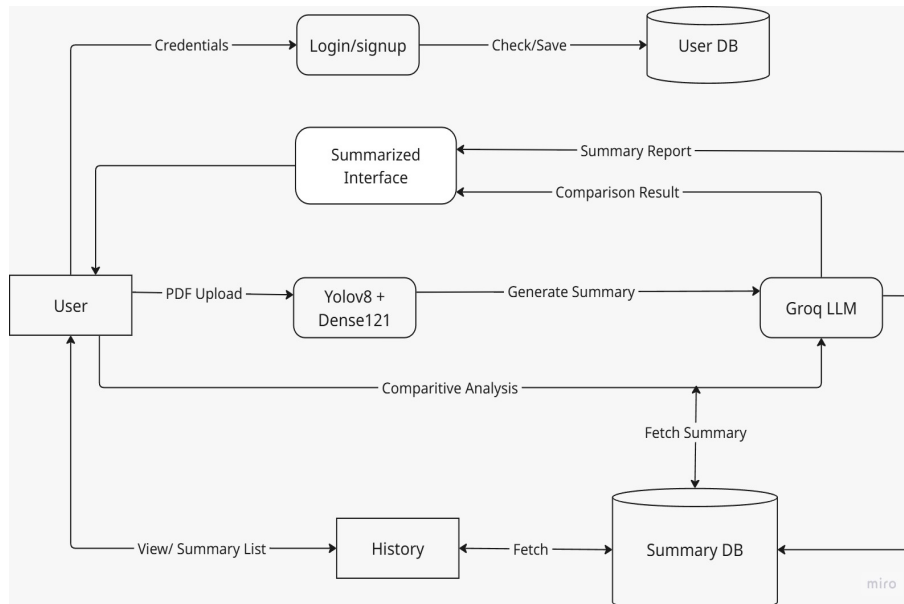


Figure 4.3: Level 1 DFD

The Level 1 DFD decomposes the system into specific functional sub-processes to illustrate the internal movement of data. User authentication is managed through the Login/Signup module, which validates and stores credentials in the User DB. The primary diagnostic flow begins with a PDF upload that is processed by the Yolov8 + DenseNet121 vision ensemble to generate report findings. These findings are then synthesized by the Groq LLM into a comprehensive Summary Report, which is presented through the Summarized Interface and persisted in the Summary DB. Additionally, the diagram highlights the Comparative Analysis workflow, where the LLM fetches

historical data from the database to generate a progress verdict, and a History module that allows users to view a list of all previously generated summaries.

4.3.3 Level 2

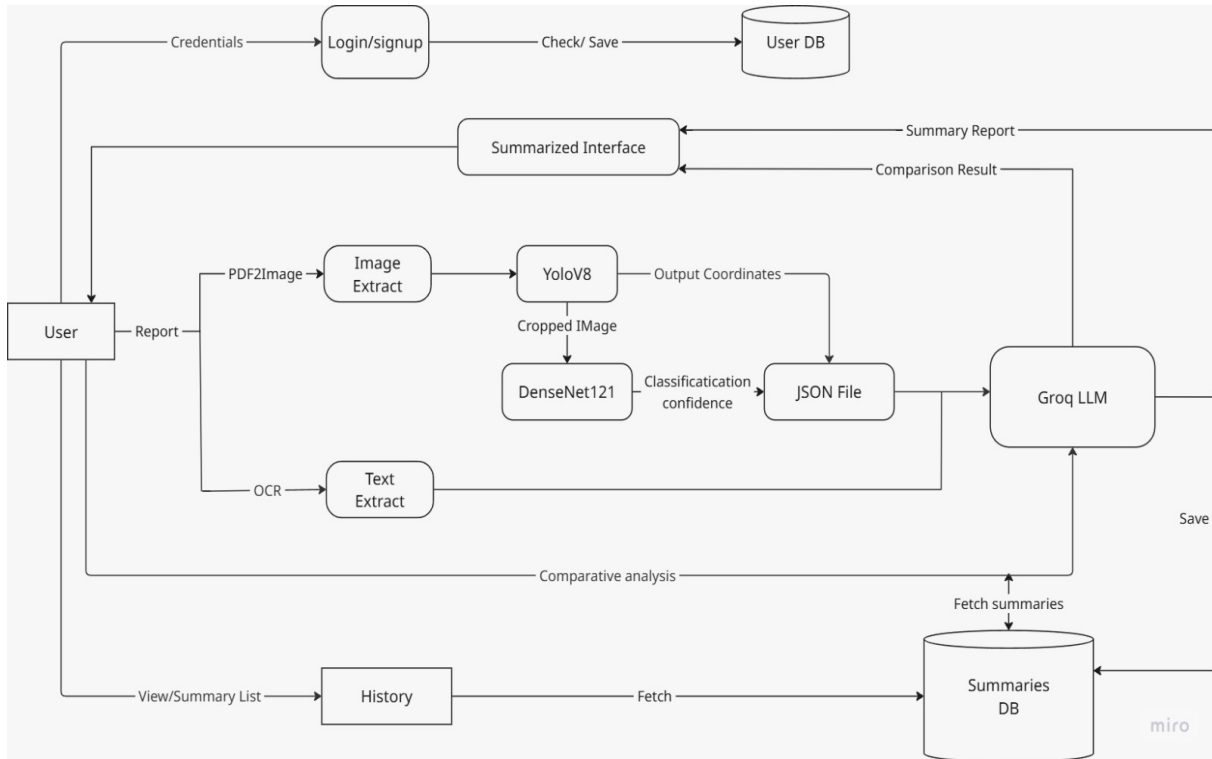


Figure 4.4: Level 2 DFD

The Level 2 DFD provides the most granular view of the system’s data transformations, detailing how raw inputs are converted into structured insights. The process begins with the Login/Signup module, where user credentials are encrypted and validated against the User DB. Upon report submission, the vision pipeline utilizes PDF2Image to convert document pages into high-resolution images for the Image Extract module. These images are processed by YoloV8, it maps precise Output Coordinates to a JSON File and generates Cropped Images of specific anomalies. These cropped segments are then analyzed by DenseNet121, which appends Classification Confidence metrics to the same JSON structure, creating a unified digital representation of the visual findings.

Simultaneously, the textual data is processed via OCR to ensure all clinical notes and vitals are captured as raw text. This text extract and the structured JSON file are then ingested by the Groq LLM, which synthesizes the multimodal data to produce a Summary Report and Comparison Result. These outputs are delivered to the Summarized Interface for user visualization while being persistently stored in the Summaries DB. For longitudinal tracking, the History module allows the user to Fetch previous summaries, while the Comparative Analysis trigger prompts the LLM to retrieve historical data to generate a clinical progress verdict.

Chapter 5

Implementation

5.1 System Requirements

5.1.1 Hardware Requirements

- **CPU:** Intel Core i5 / AMD Ryzen 7
- **GPU:** NVIDIA RTX 3060 + 8GB VRAM
- **RAM:** 16 GB

5.1.2 Software Requirements

- **Operating System:** Windows 10/11, Ubuntu 20.04
- **Core Technology:** Python 3.9
- **AI/ML Libraries:** TensorFlow, PDF2Image, Tesseract (OCR)
- **Backend:** FastAPI
- **Database:** MongoDB
- **IDE:** Visual Studio Code, Google Colab or Jupyter Notebook

5.2 Dataset Used

Medical Report Dataset:

- Patient medical reports from Kaggle

Medical Images (X-Rays) - NIH Kaggle, Roboflow

- **Training Classes:**

- COVID-19
- Normal
- Pneumonia
- Tuberculosis

- **Testing Classes:**

- COVID-19
- Normal
- Pneumonia
- Tuberculosis

5.3 Algorithms

5.3.1 Upload and Data Extraction

The user begins by uploading their medical report, typically in a PDF format, directly into the system. LucidCare then automatically performs OCR and image extraction to retrieve both the clinical text and the raw X-ray images, ensuring the user does not have to manually input complex medical data.

5.3.2 Transparent AI Analysis

During the processing phase, the system identifies abnormalities and highlights specific visual features directly on the X-ray image. This allows the user to see exactly what evidence the AI used for its conclusions, providing a clear reasoning path through Explainable AI (XAI) rather than just a final diagnosis.

5.3.3 Patient-Friendly Results

The system generates a simplified summary that translates technical radiology jargon into clear, layman-friendly language. To ensure maximum accessibility, the user can choose to view these medical insights in either English or Malayalam, which helps local patients understand their health status with confidence.

5.3.4 Health Tracking and Comparison

If the user has uploaded reports previously, the system automatically conducts a comparative analysis between the past and current results. This feature identifies specific changes in medical conditions, such as determining if parameters have increased or decreased over time, and stores this history securely for future reference.

5.4 Screenshots

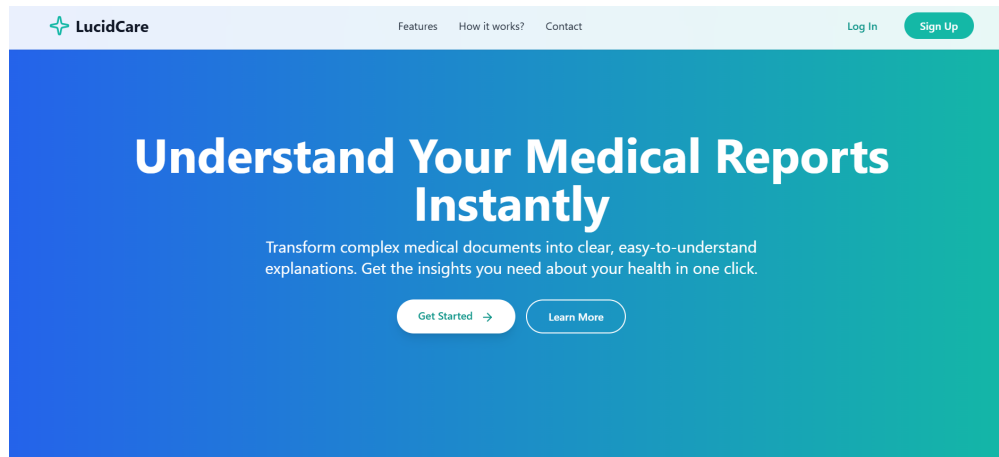


Figure 5.1: User Interface

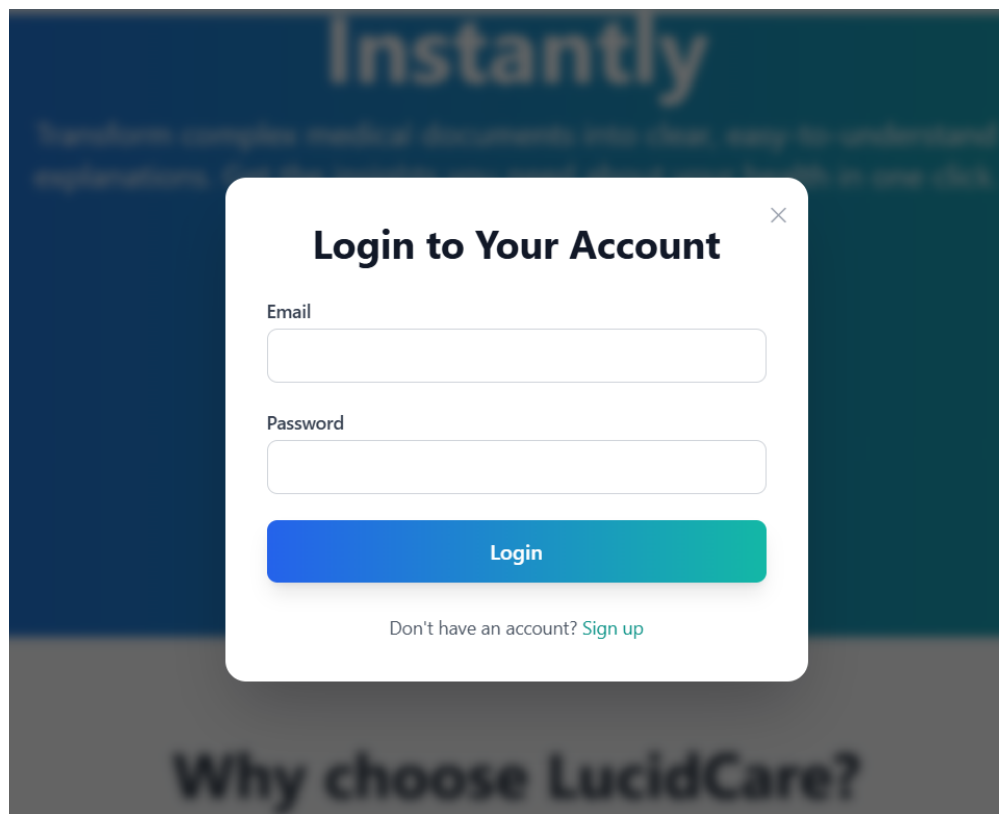


Figure 5.2: Login Page

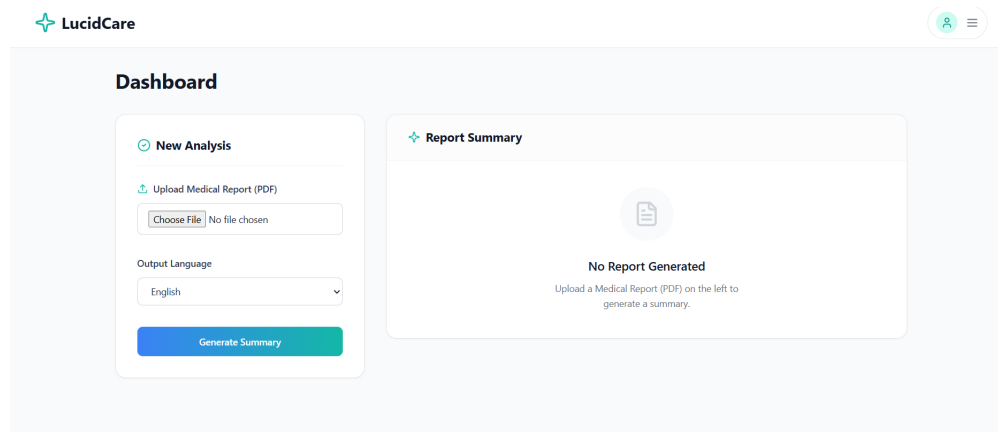


Figure 5.3: Dashboard

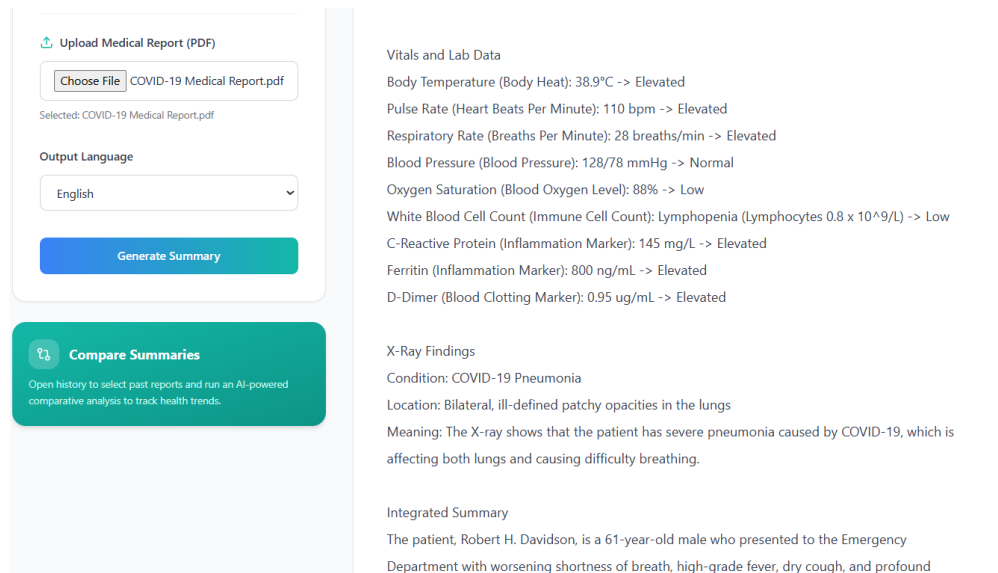


Figure 5.4: English Report

Upload Medical Report (PDF)

COVID-19 Medical Report.pdf

Selected: COVID-19 Medical Report.pdf

Output Language

Malayalam

▼

Compare Summaries

Open history to select past reports and run an AI-powered comparative analysis to track health trends.

വൈദ്യലേഖനം ലഭ്യമാണ് ഡാറ്റയും
 താപനില (ശരീര താപനില): 38.9 ഡിഗ്രി സെൽസിയസ് -> ഉയർന്നത്
 ഹൃദയസ്പന്ദനം (ഹൃദയത്തിന്റെ സ്പന്ദനം): 110 മിനിറ്റിൽ -> ഉയർന്നത്
 ശ്വാസനം (ശ്വാസനത്തിന്റെ ആവൃത്തി): 28 ശ്വാസം/മിനിറ്റ് -> ഉയർന്നത്
 രക്തസമ്മർദ്ദം (രക്തത്തിന്റെ സമ്മർദ്ദം): 128/78 മില്ലിമീറ്റർ പാസ്കൽ ->
 സാധാരണം
 ഓക്സിജൻ സാച്ചുറേഷൻ (രക്തത്തിലെ ഓക്സിജന്റെ അളവ്): 88% -> കുറവ്

എക്സ്-റേ കണ്ടെത്തലുകൾ
 അവസ്ഥ: ന്യൂമോണിയ (ശ്വാസകോശാരം)
 സ്ഥാനം: ശ്വാസകോശങ്ങൾ
 അർത്ഥം: ശ്വാസകോശത്തിലെ അണുബാധയുടെ ഫലമായി ശ്വാസകോശത്തിലെ
 എക്സ്-റേയിൽ കാണപ്പെടുന്ന അസൃഷ്ടമായ പാച്ചുകൾ.

സംയോജിത സംഗ്രഹം
 രോഗിക്ക് കോവിഡ്-19 പന്നുമോണിയ ഉണ്ടെന്ന് രോഗനിർണ്ണയം നടത്തിയിട്ടുണ്ട്.
 ശ്വാസതടസ്സം, പന്നുകളിൽ വേദന, പിണ്ഡം എന്നിവ അവതരിപ്പിച്ചു.
 ശ്വാസകോശത്തിന്റെ എക്സ്-റേയിൽ ബഹുസ്ഥല മൃദുവായ സ്ഥലങ്ങൾ
 കാണപ്പെട്ടു. രോഗിയുടെ രക്തപരിശോധനയിൽ ക്രിയാത്മക പ്രോട്ടീൻ ഉയർന്ന
 നിലയിലാണ്. രോഗിയുടെ രക്തത്തിലെ ഓക്സിജൻ സാന്നിധ്യം കുറവാണ്.

Figure 5.5: Malayalam Report

1
 Vitals and Lab Data
Today

Earlier Report

2
 Vitals and Lab Data
Today

Later Report

Condition Deteriorated
Health metrics show significant decline compared to the earlier report.

95% confidence

AI ANALYSIS SUMMARY
The patient's condition has significantly worsened, with severe acute respiratory distress syndrome, septic shock, and suspected cytokine release syndrome. The patient's vital signs and lab data indicate severe tachycardia, tachypnea, hypotension, hypoxemia, and elevated inflammation markers.

KEY METRIC CHANGES

Body Temperature **DETERIORATED**
The patient's body temperature has increased, indicating worsening infection.

38.9°C
39.8°C

Oxygen Saturation **DETERIORATED**
The patient's oxygen saturation has decreased, indicating worsening hypoxemia.

88%
74%

Figure 5.6: Compare Report

5.5 Code Snippets

5.5.1 Densenet121.py

```
# --- 3. Load DenseNet121 ---
base_model = DenseNet121(weights='imagenet', include_top=False, input_shape=(224, 224, 3))
base_model.trainable = False

x = base_model.output
x = GlobalAveragePooling2D()(x)
x = Dense(128, activation='relu')(x)
outputs = Dense(num_classes, activation='softmax')(x)

model = Model(inputs=base_model.input, outputs=outputs)
```

Figure 5.7: Densenet121 Implementation

5.5.2 YOLOv8.py

```
def convert_heatmap_to_yolo_bbox(heatmap, img_width, img_height, threshold_val):
    heatmap_resized = cv2.resize(heatmap, (img_width, img_height))
    heatmap_uint8 = np.uint8(255 * heatmap_resized)

    _, binary_mask = cv2.threshold(heatmap_uint8, threshold_val, 255, cv2.THRESH_BINARY)

    contours, _ = cv2.findContours(binary_mask, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)

    if contours:
        largest_contour = max(contours, key=cv2.contourArea)
        x, y, w, h = cv2.boundingRect(largest_contour)
```

Figure 5.8: YOLOv8 Code - Part 1

```
x_center, y_center, w_norm, h_norm = bbox
yolo_label_string = f"{yolo_class_idx} {x_center} {y_center} {w_norm} {h_norm}"

with open(output_label_path, 'w') as f:
    f.write(yolo_label_string)
```

Figure 5.9: YOLOv8 Code - Part 2

5.5.3 LLM.py

```
system_instruction = f"""
You are an expert doctor explaining a detailed diagnosis to a patient.

STRICT FORMATTING RULES:
1. PLAIN TEXT ONLY. Do not use markdown (no bold **, no headers #, no bullets -).
2. NO Emojis.
3. NO Numbered lists for sections.
4. Format exactly like the examples below.

REQUIRED OUTPUT FORMAT:

Vitals and Lab Data
[Medical Term] ([Simple Definition]): [Value] -> [Status]
[Medical Term] ([Simple Definition]): [Value] -> [Status]

X-Ray Findings
Condition: [Name]
Location: [Location]
Meaning: [Explanation]

Integrated Summary
[Detailed paragraph explaining the condition, evidence, and next steps in simple language.]

```

Figure 5.10: LLM Implementation

5.5.4 app.py

```
@app.route('/auth/login', methods=['POST'])
def Login():
    data = request.get_json()
    if not data:
        return jsonify({'error': 'Invalid JSON body'}), 400

    email = data.get('email')
    password = data.get('password')
```

Figure 5.11: Application Code

Chapter 6

Results and Discussions

6.0.1 Performance Evaluation

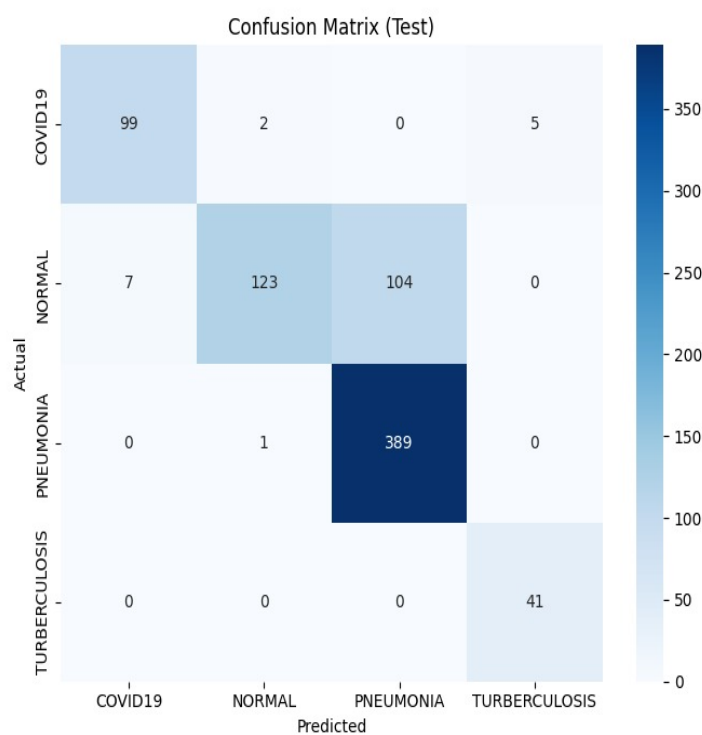


Figure 6.1: Confusion Matrix

The confusion matrix provides a granular perspective on the model's predictive performance across all four diagnostic categories, showing high accuracy along the diagonal axis where actual labels match predicted outcomes. While the model demonstrates exceptional strength in identifying COVID-19 and Pneumonia, there is a documented overlap where some normal lung scans are categorized as pneumonia, likely due to the subtle variations in lung density that characterize both states. Despite this common clinical challenge, the system achieved perfect reliability in the

detection of Tuberculosis, accurately identifying every instance of the pathology within the test set.

CLASSIFICATION REPORT (ACCURACY: 98%)

Class	Precision	Recall	F1-score	Support
COVID-19	1.00	1.00	1.00	10
Normal	1.00	0.88	0.93	8
Pneumonia	0.89	1.00	0.94	8
Tuberculosis	1.00	1.00	1.00	12
Accuracy			0.98	38
Macro Avg	0.97	0.97	0.97	38
Weighted Avg	0.98	0.98	0.98	38

Figure 6.2: Classification Report

The classification report confirms the overall robustness of the system with a total test accuracy of 98%, showcasing balanced performance across precision, recall, and F1-scores. The model achieved flawless F1-scores for both COVID-19 and Tuberculosis, while maintaining a perfect recall of 1.00 for Pneumonia to ensure that no positive cases were overlooked during the diagnostic process. Although the "Normal" category displays a slightly lower F1-score of 0.93, this is a result of the model's intentionally cautious approach, prioritizing the detection of abnormalities over the risk of occasional false positives to ensure maximum clinical safety.

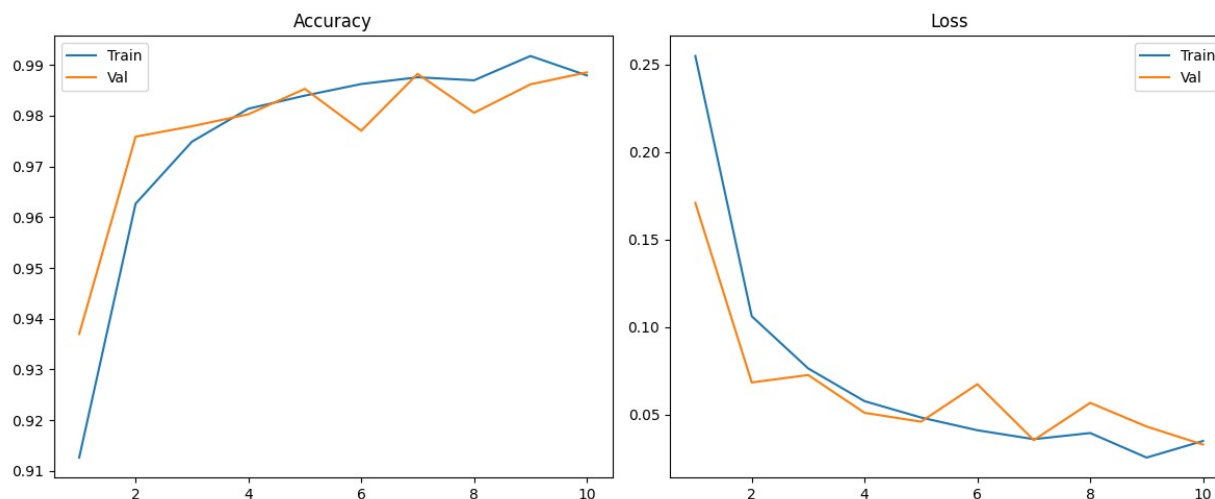


Figure 6.3: Training Accuracy and Loss Accuracy

The training and validation graphs illustrate a highly stable learning process, with accuracy for both datasets converging consistently toward the 99% threshold. Because the training and validation lines remain in close proximity throughout the 10 epochs, it is evident that the model has generalized effectively to unseen data without falling into the trap of overfitting. This successful convergence is mirrored in the loss graph, where the categorical cross-entropy loss drops sharply

from 0.25 to below 0.05, stabilizing after the seventh epoch and suggesting that the DenseNet121 architecture reached an optimal state with high efficiency.

6.0.2 Explainability Analysis

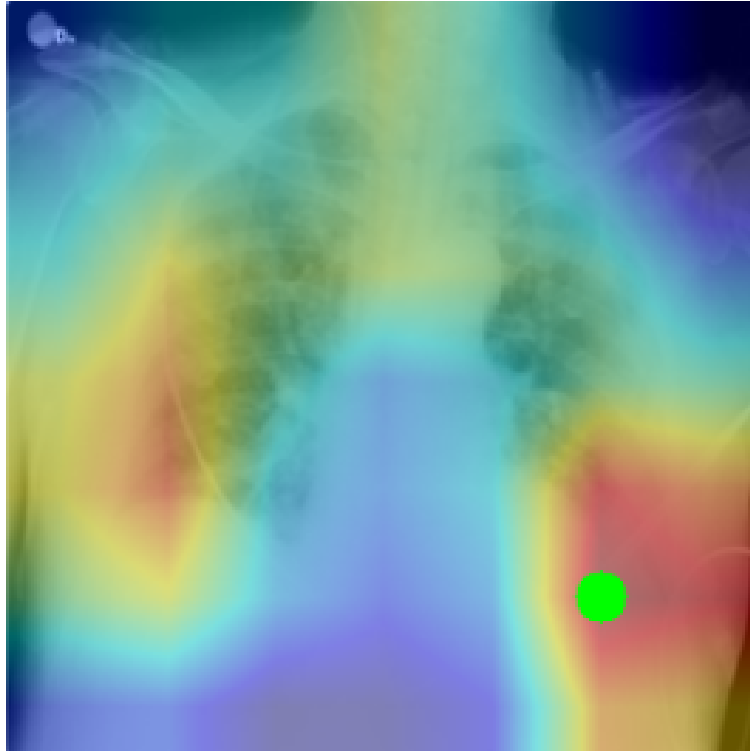


Figure 6.4: Heatmap generated by Explainable AI

The Grad-CAM overlay uses a color spectrum to highlight pathologically significant lung regions in red, providing clinical transparency for the model's decision-making process. The green hotspot identifies the peak activation point, transforming the AI from a "black box" into an interpretable tool for healthcare providers.

6.0.3 Comparision with Existing methods

Paper / Model	Approach	Key Strengths	Limitations
Çallı et al. (2022)	Explainable CNN for emphysema detection using radiologic signs	High interpretability using explicit clinical features, strong alignment with radiologist reasoning	Limited to a single disease, requires annotated radiologic signs, no text integration
Anderson et al. (2024)	Deep learning assisted diagnosis for chest X ray interpretation	Improves physician accuracy, strong clinical validation through reader studies	No patient facing explanation, no multimodal integration, depends on clinical setting
Sun et al. (2023)	Multimodal deep learning review (image and text)	Highlights importance of multimodal fusion and improved performance	No implementation, lacks real system validation, theoretical insights only
Fi et al. (2024)	Retrieval augmented generation for biomedical text summarization	Improves readability and accessibility of medical text, reduces hallucination	No image understanding, limited to text summarization, no clinical visual reasoning
Naz et al. (2023)	CNN with explainability for pulmonary disease classification	Combines classification with explainability, improves model transparency	Uses post hoc explanations which can be unstable, no multimodal fusion
Park et al. (2024)	LLM based patient friendly radiology report generation	Enhances patient understanding through simplified summaries	No image analysis, risk of hallucination, limited clinical grounding
Ji et al. (2024)	Vision language model for radiology report generation	Integrates image and text for report generation, strong multimodal capability	Dependent on dataset quality, lacks explainability, not patient focused
Proposed Model (LucidCare)	Hybrid multimodal system combining YOLOv8, DenseNet, and LLM with explainable AI	Integrates image and text data. Provides visual and textual explainability. Generates patient friendly summaries. Supports bilingual output and comparative analysis.	Limited dataset diversity. Requires further clinical validation

Table 6.1: Comparison of Existing Models with Proposed System

Chapter 7

Conclusion

Our project introduces an AI-driven system designed to make complex radiology reports clear, visual, and easily understandable for everyone, regardless of their medical background. By combining a YOLOv8 and classifier ensemble for detailed image analysis with a Large Language Model (LLM) for natural language explanation, the system produces accurate, patient-friendly reports that are enhanced with explainable AI components. This ensures that users not only receive diagnostic results but also understand the reasoning behind them through visual highlights and simplified textual explanations.

A defining feature of this system is its bilingual reporting capability. Reports are generated in both English and Malayalam, ensuring that critical medical information is accessible to local communities and not limited by language barriers. This feature empowers patients by enabling them to better understand their health conditions and make informed decisions with confidence. Through this approach, the system bridges the gap between advanced medical technology and patient accessibility, promoting inclusivity and health awareness at the community level.

Chapter 8

Future Scope

In future iterations, LucidCare aims to incorporate enhanced data security and compliance mechanisms, including robust role-based access control, encrypted data handling, and comprehensive audit logging to align with evolving healthcare privacy standards. These improvements will ensure greater trust, accountability, and safe management of sensitive patient information.

The system is also intended to be further developed as a clinical decision-support and second-opinion tool, explicitly avoiding full automation in diagnosis. By strengthening its assistive capabilities, LucidCare will support clinicians in making informed decisions while preserving human oversight and clinical responsibility.

Additionally, future enhancements will focus on integrating risk stratification and case prioritization features. This will enable the system to categorize patients based on severity and urgency, helping healthcare professionals efficiently allocate attention to critical cases and improve overall clinical workflow management.

Appendix A

Bibliography

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Appendix B

Publications

B.1 PAPER STATUS

Paper Title: LucidCare: An Explainable Multimodal AI System for Patient-Centric Clinical Report Interpretation

Paper ID: 130

Conference: 2026 IEEE 5th International Conference on Signal Processing, Informatics, Communication and Energy Systems (IEEE SPICES2026)

Organised at: Government College of Engineering, Kannur

Publication Status: Waiting for Approval

Paper Title: LucidCare: An Explainable Multimodal AI System for Patient-Centric Clinical Report Interpretation

Paper ID: 220

Conference: International Conference on Smart Communication and Sustainable Technologies (IC-SCST 2026)

Organised at: Saintgits College of Engineering

Publication Status: Paper Accepted (Not yet published)